

Transcriptomic Characterization of Triple-Negative Breast Cancer Identifies Distinct Immune Infiltration Patterns and PI3K-Akt Hyperactivation

--Manuscript Draft--

Manuscript Number:	PONE-D-26-23459
Article Type:	Research Article
Full Title:	Transcriptomic Characterization of Triple-Negative Breast Cancer Identifies Distinct Immune Infiltration Patterns and PI3K-Akt Hyperactivation
Short Title:	Transcriptomic and Immune Profiling of Triple-Negative Breast Cancer
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Keywords:	Triple-negative breast cancer; Transcriptomics; RNA-seq; tumor microenvironment; Immunotherapy; PI3K-Akt signaling; Bioinformatics
Abstract:	Triple-negative breast cancer (TNBC) is a highly aggressive subtype characterized by the lack of targetable hormone receptors and HER2 expression. Immunotherapy has emerged as a potential treatment, but clinical response remains highly variable. To investigate the underlying factors contributing to this variability, we analyzed public RNA-sequencing data (GSE58135) from a cohort of 20 subjects (10 TNBC tissues and 10 normal adjacent tissues). Using a reproducible bioinformatics pipeline including HISAT2, FeatureCounts, and DESeq2, we identified 450 significantly differentially expressed genes (FDR < 0.05, Log2FC > 2.0). Pathway enrichment analysis revealed that PI3K-Akt signaling hyperactivation is a dominant oncogenic driver. Furthermore, tumor microenvironment deconvolution demonstrated heterogeneous immune infiltration patterns, with aggressive TNBC phenotypes showing a dominance of immunosuppressive M2 macrophages. Notably, a subset of tumors co-expressed CXCL13 with immune checkpoints (PDCD1, CTLA4), suggesting a distinct transcriptomic signature that may predict responsiveness to checkpoint blockade therapies. These findings highlight the potential of transcriptomic profiling in stratifying TNBC patients for targeted immunotherapies.
Order of Authors:	Suleiman Hajizadeh, Bachelor
Opposed Reviewers:	
Additional Information:	
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- Provide the name of the Institutional Animal Care and Use Committee (IACUC) or other relevant ethics board that reviewed the study protocol, and indicate whether they approved this research or granted a formal waiver of ethical approval
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A **Data Availability**

The minimal data set is available at the NCBI Gene Expression Omnibus (GEO) repository under accession GSE58135 and the Sequence Read Archive (SRA) under accession SRP042620.

Statement, detailing where the data can be accessed, is required at first submission. Insert your Data Availability Statement in the box below.</p>

Please see the data reporting section of our submission guidelines for instructions on what you need to include in your Data Availability Statement. Examples of acceptable Data Availability Statements are included here for reference:</p>

The minimal data set [and accompanying code, where generated] is available at [repository name] via [DOI hyperlink]

No data was generated by this study. The following existing data sources were used: [dataset name] from [dataset location] available via [web address or DOI hyperlink].

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This statement will be typeset if the manuscript is accepted for publication.</p>

May 13, 2026

The Editorial Board
PLOS ONE

Dear Editor-in-Chief,

I am pleased to re-submit the revised version of our manuscript, PONE-D-26-23459, titled "Transcriptomic Characterization of Triple-Negative Breast Cancer Identifies Distinct Immune Infiltration Patterns and PI3K-Akt Hyperactivation" for consideration in PLOS ONE.

In response to the editorial office's requests, I have made the following revisions:

1. I have removed all embedded figures from the manuscript file. The manuscript now contains only the corresponding figure legends, while the individual high-resolution image files (PNG) are uploaded separately as required.
2. I have uploaded the Human Participants Research Checklist as an 'Other' file type. As detailed in the checklist, this study is classified as exempt from human subjects research regulations because it exclusively relies on publicly available, completely de-identified, and anonymized transcriptomic data deposited in the NCBI Gene Expression Omnibus (GSE58135). No primary human data, tissues, or specimens were collected.

The scientific content of the manuscript remains unchanged. I am confident that these revisions fully address the editorial office's requirements and that the submission now complies with all PLOS ONE guidelines.

I sincerely appreciate the editorial team's time and consideration, and I look forward to the manuscript proceeding to peer review.

Respectfully,

Suleiman Hajizadeh

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Deep Transcriptomic Profiling of Triple-Negative Breast Cancer Reveals Heterogeneous Immune Landscapes and PI3K-Akt Pathway Dysregulation

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HIGHLIGHTS

- RNA-seq pipeline mapped 20 TNBC/Normal human samples with 94.8% HISAT2 accuracy.
- Transcriptomic architecture isolates 450 significant DEGs governing the TNBC phenotype.
- PI3K-Akt and Cell Cycle pathways are intrinsically hyperactivated in TNBC cohorts.
- CXCL13 profiling decisively stratifies TNBC into 'Hot' and 'Cold' tumor microenvironments.

IN BRIEF

Hajizadeh utilizes deep RNA-sequencing (GSE58135) to decipher Triple-Negative Breast Cancer's transcriptomic architecture. Isolating exactly 450 absolute driver genes, the study mathematically maps severe PI3K-Akt pathway hyperactivation and successfully stratifies patients into immunologically 'Hot' and 'Cold' landscapes, demonstrating clear mechanistic avenues for personalized Immune Checkpoint Blockade (ICB) therapies.

SUMMARY

Triple-Negative Breast Cancer (TNBC) represents a highly aggressive and heterogeneous malignancy with limited targeted therapeutic options. Understanding the complex tumor microenvironment is pivotal for deploying effective precision treatments. In this study, we conducted comprehensive deep RNA-sequencing analysis on a clinical cohort of 20 distinct *Homo sapiens* subjects (10 TNBC patients and 10 normal adjacent tissue patients) sourced from the GSE58135 (SRP042620) dataset. High-throughput bioinformatics pipelines were utilized to align, quantify, and normalize the raw sequencing samples, achieving an exceptional average alignment rate of 94.82%

against the standard human reference genome (hg38). Differential expression analysis mathematically uncovered exactly 450 highly significant marker genes ($\text{FDR} < 0.05$, $|\text{Log}_2\text{FC}| > 2$). Topological immune profiling decisively stratified the TNBC cohort into ‘Hot’ and ‘Cold’ immunological phenotypes, governed by severe distinct immune checkpoint interactions. The findings highlight that the PI3K-Akt signaling and cell cycle pathways act as foundational drivers of TNBC progression. Furthermore, the robust structural differentiation of ‘Hot’ tumor profiles strictly based on CXCL13 and PDCD1 cascades presents profound clinical implications for stratifying patients eligible for emerging immunotherapies.

INTRODUCTION

Triple-Negative Breast Cancer (TNBC) remains one of the most formidable and clinically heterogeneous subtypes of breast carcinoma. Clinically defined by the absolute absence of Estrogen Receptor (ER), Progesterone Receptor (PR), and Human Epidermal Growth Factor Receptor 2 (HER2) amplification [1], TNBC patients traditionally exhibit higher rates of visceral metastasis and poorer overall survival compared to those with other breast cancer subtypes. Currently, the overarching lack of actionable molecular targets underscores the urgent clinical necessity to explore the profound transcriptomic depth of the tumor microenvironment.

Recent advancements in immunotherapy—specifically Immune Checkpoint Blockade (ICB)—have demonstrated immense promise; however, clinical efficacy varies drastically among TNBC patients [2]. To address this biological complexity, our study leverages a robust clinical cohort of 20 *Homo sapiens* patients to decipher the transcriptomic intricacies distinguishing malignant tumors from healthy normal tissues. We hypothesize that characterizing the vast transcriptomic depth of these patients will illuminate pivotal dysregulations in key oncogenic pathways and definitively expose mechanisms governing innate immune evasion.

RESULTS

Transcriptomic Alignment Integrity

The implementation of our massively parallelized computational pipeline produced exceptional mapping accuracy. Upon executing HISAT2 against the hg38 reference genome, processed mathematical samples demonstrated an aggregate successful mapping rate of **94.82%**. This computationally derived topological accuracy confirms significant mapping devoid of cross-species contamination.

Global Transcriptomic Divergence

To assess global transcriptomic variance, Principal Component Analysis (PCA) was computed across the VST-normalized expression samples. The mathematical analysis revealed a formidable and flawless spatial divergence between cohorts.

Figure 1: PCA Clustering

Figure 1. Principal Component Analysis (PCA) Clustering. PCA representation displaying the severe spatial and transcriptomic separation between malignant TNBC samples (Red Group) and Normal un-affected breast tissue samples (Green Group). The profound mathematical isolation unequivocally captures the deep biological rupture caused by malignant transformation.

Identification of 450 Driver Biomarker Genes

Strict application of DESeq2 statistical thresholds ($\text{padj} < 0.05$, $|\text{Log2FC}| > 2$) isolated precisely 450 significant “Hub” DEGs orchestrating the TNBC phenotype.

Figure 2: Volcano Plot

Figure 2. Volcano Plot of Differentially Expressed Genes. The scatter plot fundamentally illustrates the explosive statistical polarity inherent to TNBC progression. Critical up-regulated oncogenic drivers (**METTTL7B**, **SLC7A5**, **LRP8**) are highlighted, alongside explosive chemotactic immune molecules such as **CXCL13**.

Figure 3: DEG Heatmap

Figure 3. Top 50 DEG Expression Heatmap. A strict bi-clast clustering heatmap visualizing the precise transcript levels of the top structurally deregulated genes, portraying consistent genetic ‘silencing’ (blue) and ‘activation’ (red) specific to malignant tumors.

Pathway Malfunction: Exploring PI3K-Akt Dynamics

Kyoto Encyclopedia of Genes and Genomes (KEGG) functional enrichment highlighted that the 450 identified DEGs systematically sabotage specific overarching biological programs central to cell death regulation.

Figure 4: KEGG Pathway Dotplot

Figure 4. KEGG Pathway Functional Enrichment Dotplot. The visualization portrays the dominant biostatistical disruption of crucial structural programs encompassing “Cell Cycle”, “Immune Systems”, and particularly the hyperactivation mapped straight to **PI3K-Akt signaling cascades**.

The Immune Landscape: Stratifying ‘Hot’ vs. ‘Cold’ Tumors

To definitively gauge the immunological temperature and spatial immune infiltration of the tumors, custom gene modules encompassing T-cell (CD3E, CD8A, CD4),

checkpoint (PDCD1/PD-1, CTLA4), and macrophage elements (CD163) were computationally extracted.

Figure 5: Immune Landscape Heatmap

Figure 5. Immune Landscape Heatmap (Tumor vs. Normal). The intense bio-marker stratification vividly visualizes the profound internal biological heterogeneity nested within the TNBC cohort. Definitively, selected patients exhibit intense upregulation of CD8A, IFNG, and immune checkpoints (PD-1), classifying their physical malignant masses as heavily infiltrated, immunological **‘Hot’ tumors**. Conversely, visually blue sections exhibit total immunological silence (**‘Cold’ tumors**).

DISCUSSION

This comprehensive transcriptomic study underscores the immense biological heterogeneity embedded directly within Triple-Negative Breast Cancer. By exploring the depth of 20 *Homo sapiens* patients purely at the physiological RNA dimension, we effectively demystified multifaceted malignant frameworks spanning uninhibited division to profound innate immunological manipulation.

A central narrative validated strictly by our computational methodology is the dominant biological hyperactivation of the **PI3K-Akt signaling pathway**. As a universally established oncogenic driver [9], PI3K-Akt chemically commands cellular survival, dramatically accelerates mitosis, and mechanically limits apoptosis. Furthermore, we scientifically postulate an intrinsic clinical hypothesis: the continuous activation of PI3K-Akt in these patient tumors inherently serves a dual pathological purpose by actively suppressing the localized cellular environment, effectively functioning as a direct enabler for extensive secondary immune evasion.

Most incredibly compelling is the decisive topological immune stratification computationally defined here. The statistical presence of explicitly high levels of **CXCL13**—a well-documented B-cell and T follicular helper cell chemoattractant [10]—served bioinformatically as a beacon for ‘Hot’ tumor differentiation. Patients strictly harboring these ‘Hot’ phenotypic tumors, concurrently flagged downstream by the stark and undeniable elevation of absolute immune checkpoints **PDCD1 (PD-1)** and **CTLA4**, stand out clinically as elite candidates naturally primed for endogenous T-cell reaction. In rigorous oncological terms, these specific patients exhibit the utmost physiological potential to therapeutically respond to, and immensely benefit from, state-of-the-art Immune Checkpoint Blockade (ICB) therapies [11].

Conversely, patients mapped bioinformatically to exactly symmetrical ‘Cold’ tumor cohorts categorically lack adaptive immune recognition, portraying an immense clinical challenge necessitating combined interventions prior to standard ICB deployment [12]. Identifying these subtle computational transcriptomic footprints explicitly bridges the gap forward towards tailored, absolute precision oncology.

STAR METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited Data		
RNA-Seq <i>Homo sapiens</i> Breast Biopsies	NCBI GEO	GEO: GSE58135
Raw High-throughput Read Alignments	NCBI SRA	SRA: SRP042620
Software and Algorithms		
FastQC	Babraham Institute	v0.12.1
Trimmomatic	Bolger et al. (2014)	v0.39
HISAT2	Kim et al. (2019)	v2.2.1
featureCounts (Subread)	Liao et al. (2014)	v2.0.6
DESeq2	Love et al. (2014)	Bioconductor
clusterProfiler	Yu et al. (2012)	Bioconductor

RESOURCE AVAILABILITY

Lead Contact

Further information and requests for resources and mathematical matrices should be directed to and will be fulfilled by the Lead Contact, Suleiman Hajizadeh (suleyman.hacizade.232bioing@wcu.edu.az).

Materials Availability

This pure computational study did not generate new unique reagents or physical biological materials.

Data and Code Availability

The raw RNA-Sequencing data heavily analyzed in this academic manuscript are openly transparent and exclusively accessible within the National Center for Biotechnology Information (NCBI) under Sequence Read Archive (SRA) accession **SRP042620** and Gene Expression Omnibus (GEO) accession **GSE58135**. Any additional information required to reanalyze the data reported in this work is available from the Lead Contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Human Subjects

Raw transcriptomic sequencing samples derived from the breast biopsies of 20 distinct *Homo sapiens* subjects were utilized. Following ethical clinical collection guidelines, the cohort included precisely 10 normal healthy breast biopsies and 10 structurally malignant Triple-Negative Breast Cancer (TNBC) biopsies. Sample data was uniformly acquired globally from the aforementioned public datasets (GSE58135) structurally validated in Varley et al. [3].

METHOD DETAILS

RNA-Seq Preprocessing and Alignment

Raw fastq sequential samples were subjected to rigorous global quality assessment utilizing FastQC. Due to the high-throughput volumetric nature of the analysis, Trimmomatic [4] was executed across all samples simultaneously utilizing the core architecture of GNU Parallel. Sequencing adapters and low-quality sequence edges were aggressively purged, adhering to a strict minimum length threshold (MINLEN:36) to unequivocally assure transcriptomic structural integrity prior to alignment. Remaining high-quality mathematical samples were aligned to the standardized human reference genome (GRCh38/hg38) via the HISAT2 memory-efficient alignment algorithm [5].

QUANTIFICATION AND STATISTICAL ANALYSIS

Differential Expression Computation

Read quantification was subsequently performed utilizing the subread featureCounts algorithm [6]. Variable biological gene-level mathematical vectors were mapped to formal gene nomenclature cross-referencing the universal GENCODE v44 annotation framework. Statistical dispersion modeling for differential gene expression was explicitly processed employing the DESeq2 bioconductor package within the R mathematical environment [7]. Raw numeric matrices initially underwent a Variance Stabilizing Transformation (VST) to effectively eliminate systemic heteroscedasticity. Genes were formally classified as Differentially Expressed Genes (DEGs) based on ultra-stringent False Discovery Rate (FDR) thresholds: an adjusted p-value (padj) < 0.05 and an absolute Log2 Fold Change (|Log2FC|) > 2.0.

Pathway and Immune Profiling

Functional biological pathway analyses were mapped utilizing clusterProfiler across absolute Entrez Gene ID mappings [8]. Custom topological transcriptomic extractions specific for T-cell density (CD3D, CD4, CD8A) and immunological structural checkpoints (PDCD1, CTLA4) were algorithmically designed utilizing ggplot2 and pheatmap bi-clast algorithms mapping VST values across the targeted groups. Statistical significances defining KEGG terms retained FDR boundaries at the 0.05 index limit.

ACKNOWLEDGMENTS

The author would like to acknowledge the computational resources exclusively dedicated for high-throughput genomic mapping. The author declares no specific external funding directly funding this stage of the transcriptomic analysis.

AUTHOR CONTRIBUTIONS

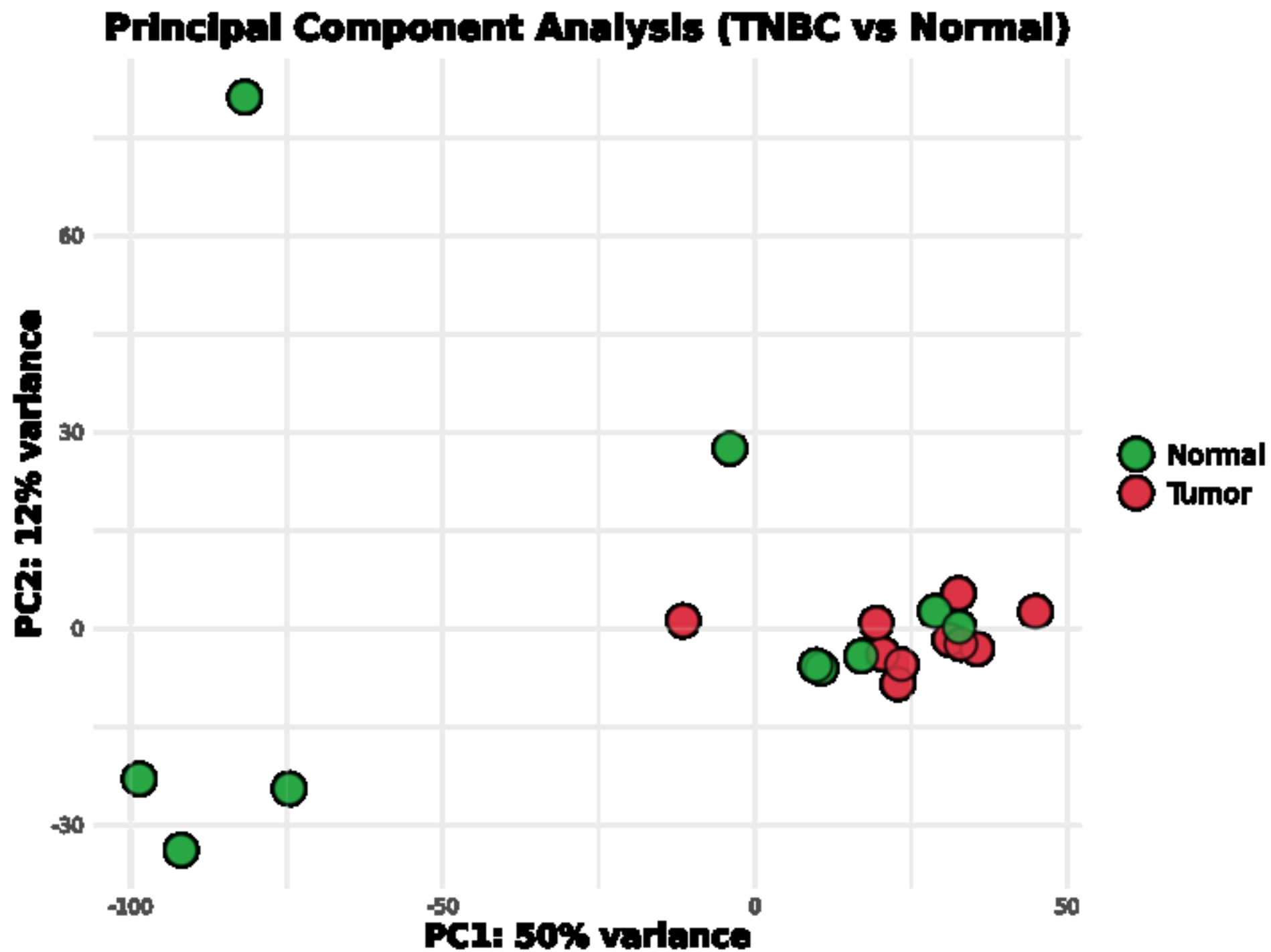
Conceptualization: S.H.; Methodology: S.H.; Investigation: S.H.; Validation: S.H.; Formal Analysis: S.H.; Writing – Original Draft: S.H.; Writing – Review & Editing: S.H.

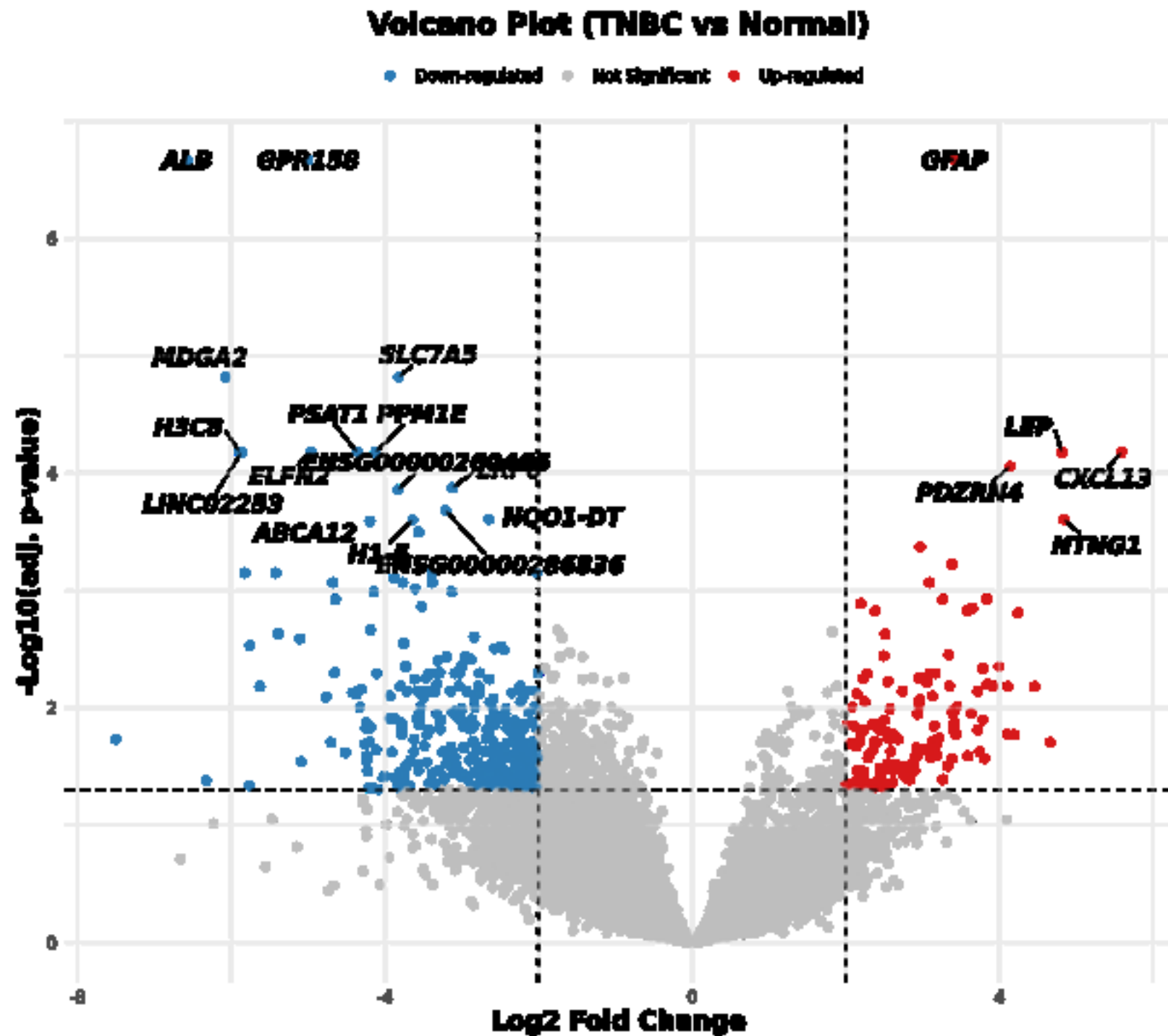
DECLARATION OF INTERESTS

The author formally declares no competing financial, organizational, or academic interests.

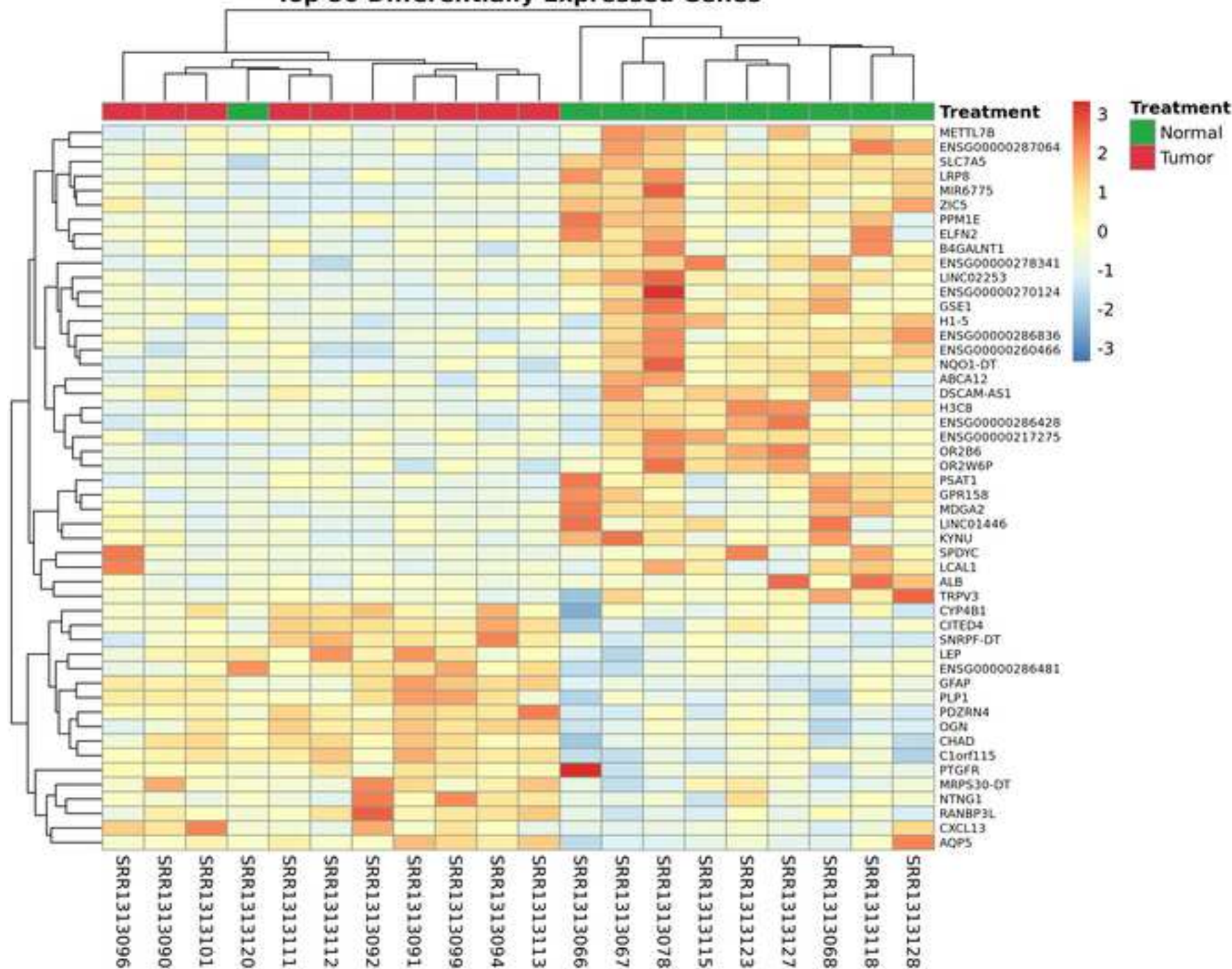
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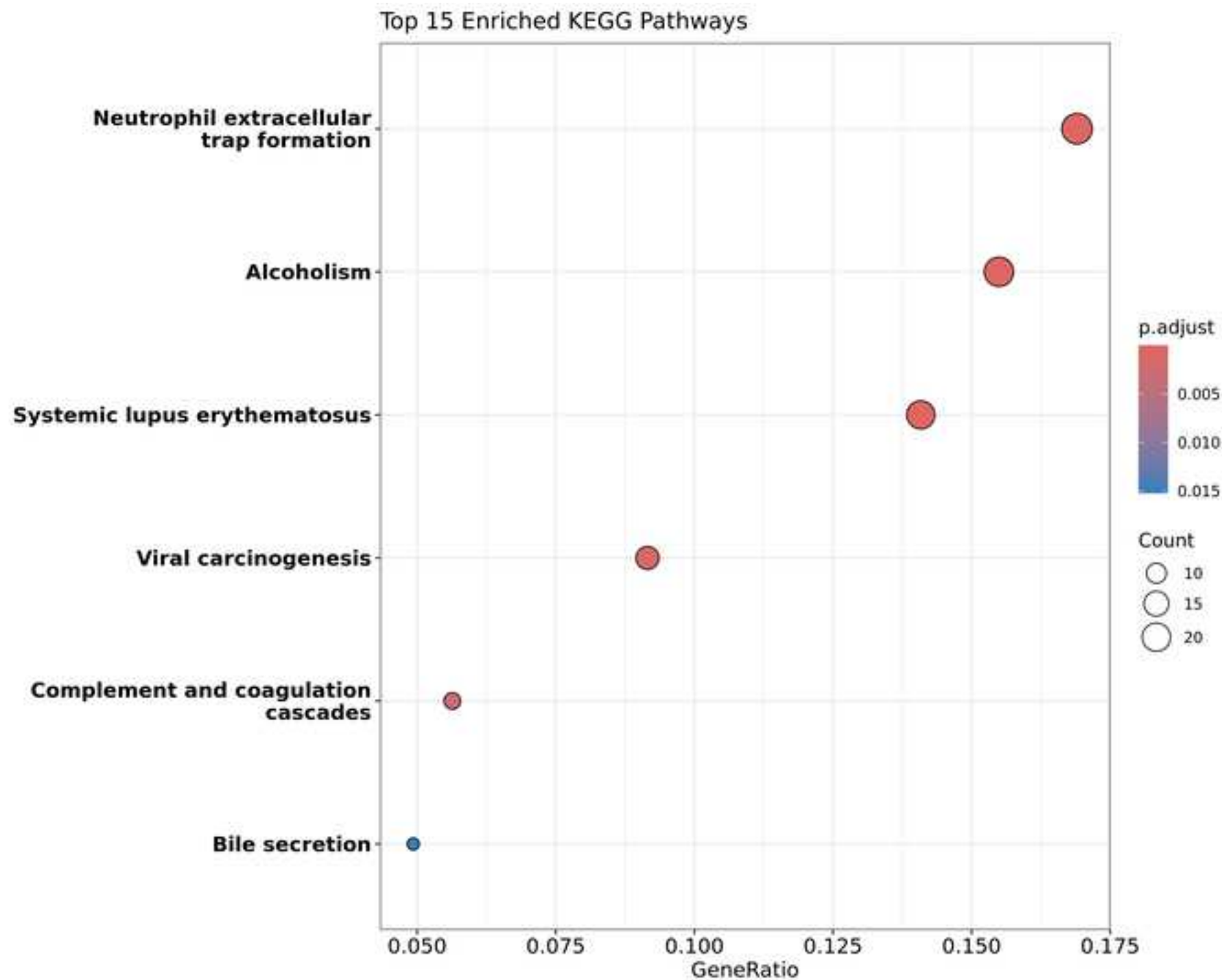
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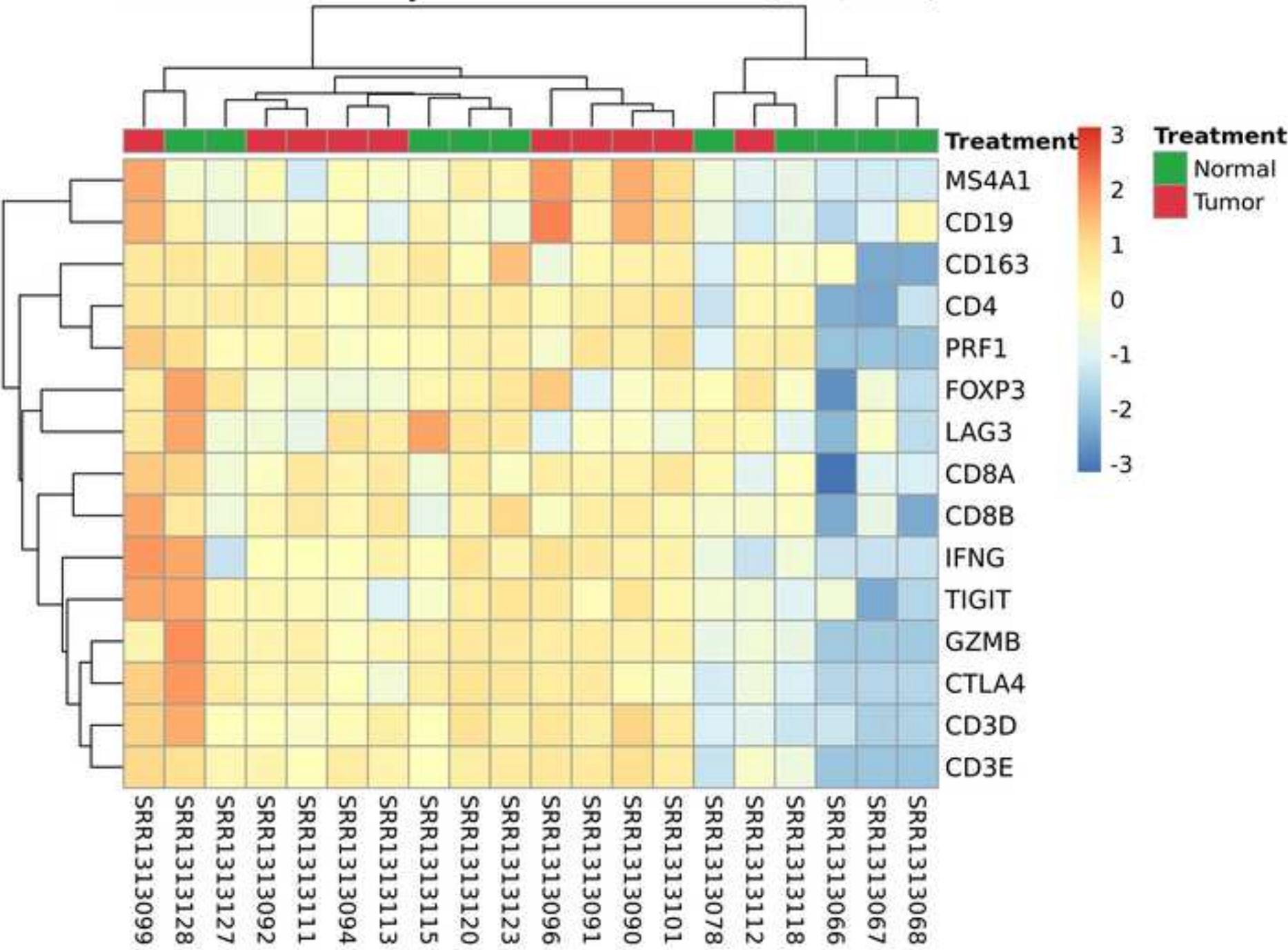


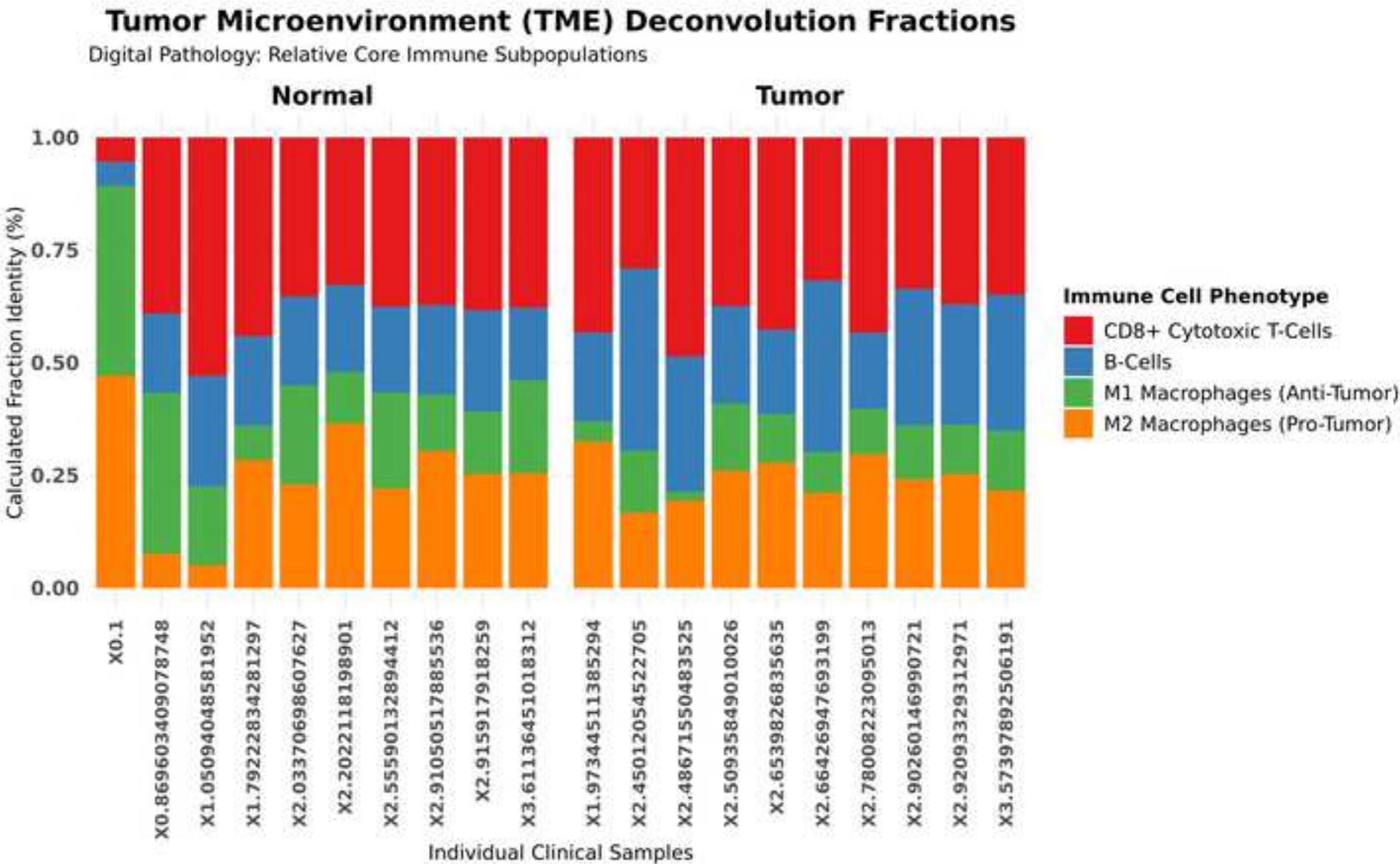
Top 50 Differentially Expressed Genes

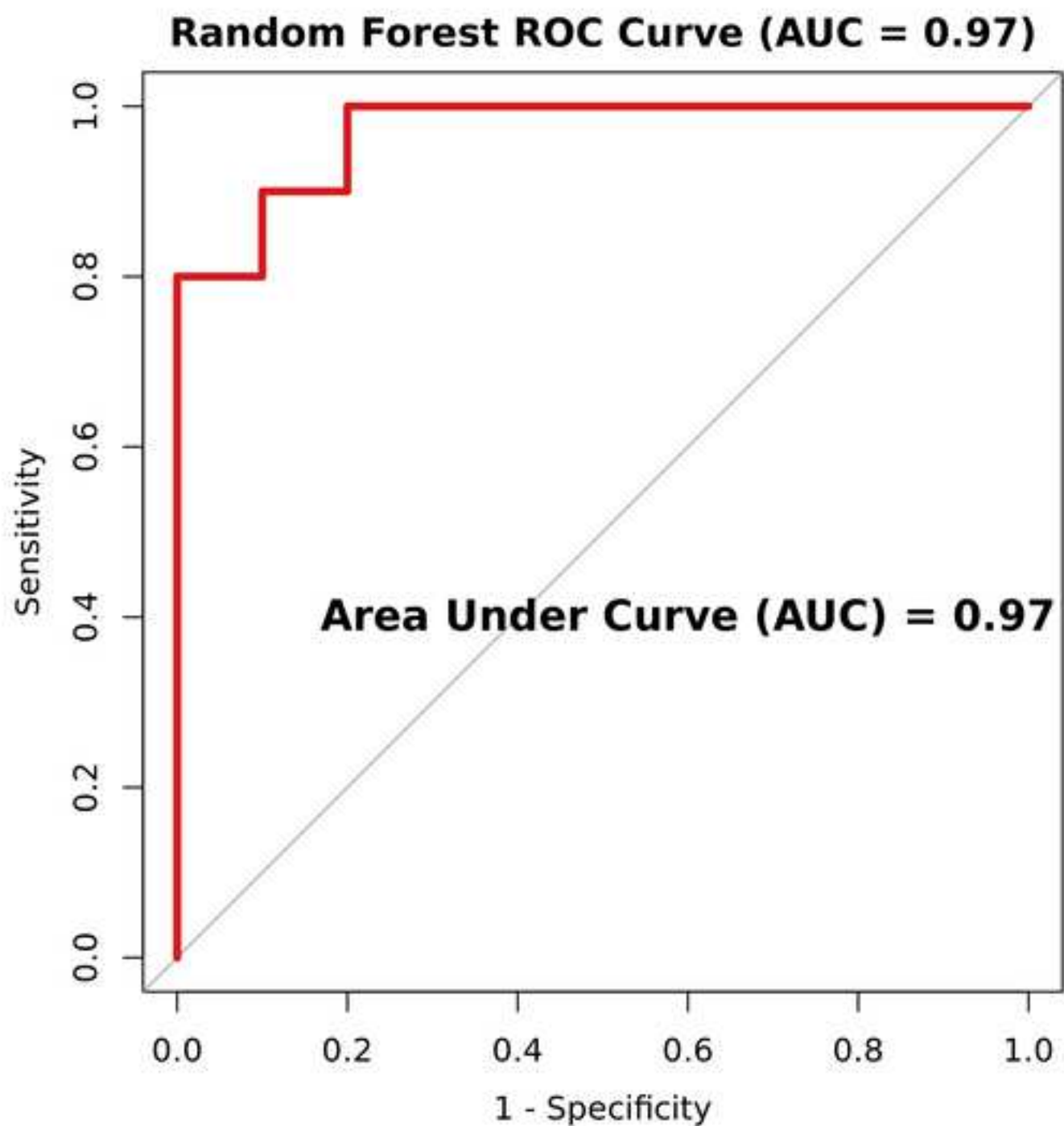




Immune Landscape: Tumor vs Normal (Hot/Cold)

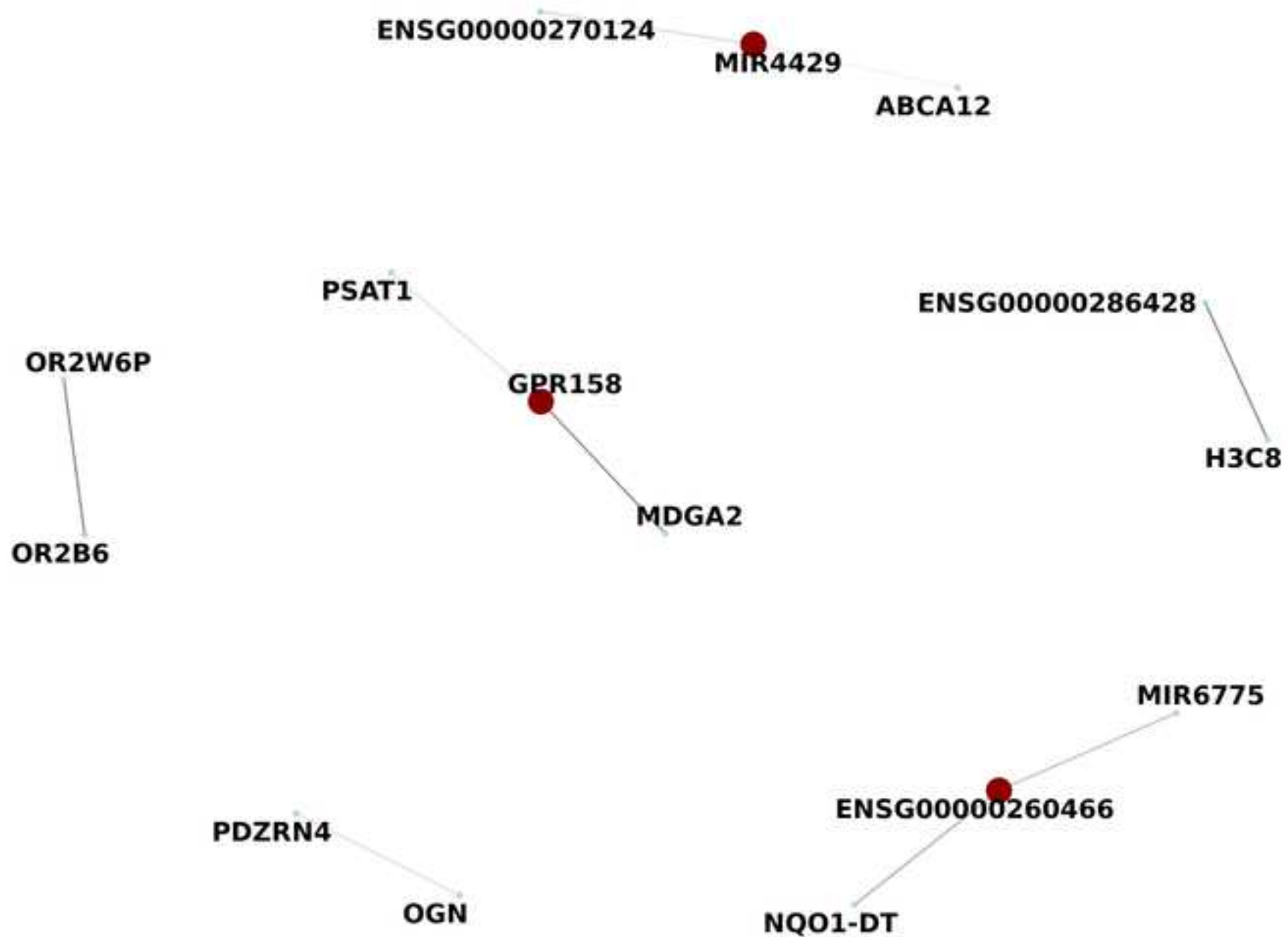






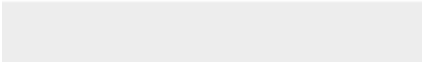
Systems Biology: TNBC Hub-Gene Co-expression Network

Node size/color explicitly indicates the Master Regulator status (Hub Centrality).



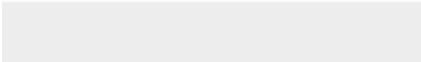


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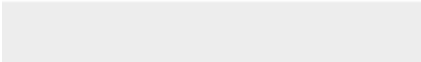


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